

Electricity Demand Forecasting Using AI

Project Lab Automation

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December 18, 2025

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Data retrieval updated to use only API

Trained baseline model and tested different regressor with an autoregressive model

Open tasks

- Incorporate data from the equivalent date in previous year(s)
- Interpret results and check feature importance

Seasonal features

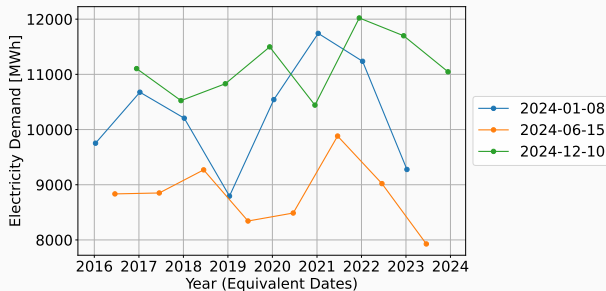


Figure 1: Seasonal features.

Electricity demand at weekday-aligned equivalent dates shows high variability across years and reference dates

→ Limited additional predictive value of these seasonal features

Correlation

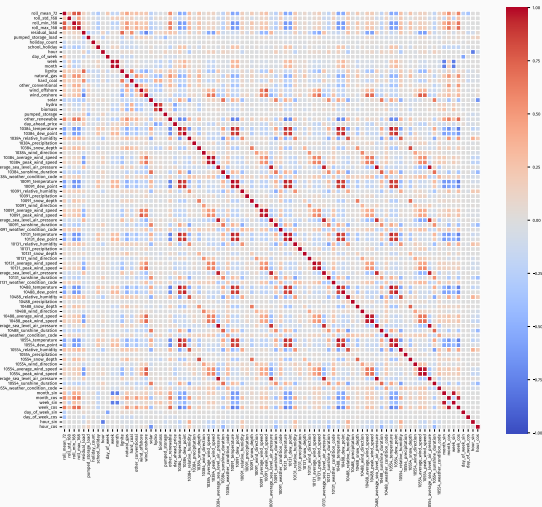


Figure 2: Correlation of exogenous features.

Feature Importance – Exogenous Variables

Quantifies how much a feature contributes to reducing the model loss

Two complementary measures are considered:

- **Split importance:** number of times a feature is used for splitting
- **Gain importance:** total reduction of the loss function achieved by splits using the feature

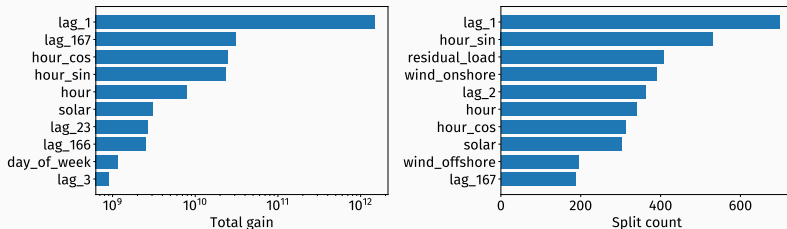


Figure 3: Feature importance of exogenous variables.

Feature Importance – Lags

Lagged target values dominate the predictive performance

Important Note: y-axis in the gain plot is logarithmic!

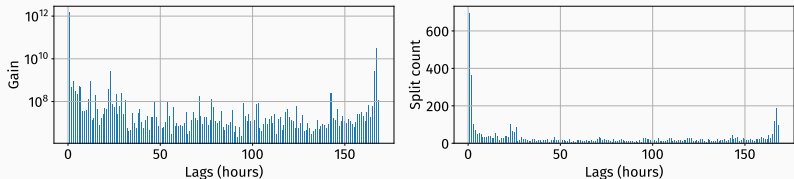


Figure 4: Feature importance of lags.

SHAP values

SHAP values explains individual model predictions [2]

Decomposes a prediction into additive contributions of each feature

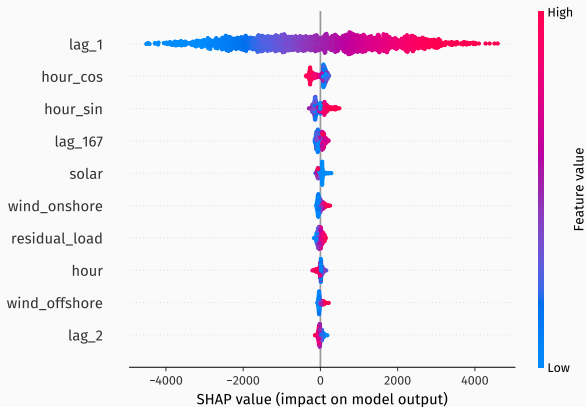


Figure 5: SHAP summary.

- **Partial Dependence Plots (PDP):**
Show the average effect of a feature on the model prediction
- **Individual Conditional Expectation (ICE):**
Visualize feature effects for individual samples
- **Permutation Feature Importance:**
Measure performance degradation when a feature is randomly permuted

Road Map

1. Create a dataset ☒
2. Graphical exploration ☒
3. Comparing and choosing the best model ☒
4. Training and backtesting ☒
5. Exogenous Variables ☒
6. Feature selection (explanability and interpretability) ⌚
7. Iterate through 3.-6. until satisfied with the results ☒
8. Deep Learning approach, Probabilistic forecasting ⊗

Gradient Boosting Machine

Algorithm [1]:

1. Start with simple model, e.g. mean of target variable

$$F_1(x) = f_1(x) = \bar{y}$$

2. Calculate error gradient

$$\hat{r}_{mi} = - \left. \frac{\partial \mathcal{L}(y_i, F(x_i))}{\partial F(x_i)} \right|_{F=F_m} \quad \forall i \in \{1, \dots, N\}$$

3. Fit a new weak learner to the residuals

$$f_{m+1}(x) \approx \hat{r}_m$$

4. Determine how much of the new learner to add

$$\gamma_{m+1} = \arg \min_{\gamma} \sum_{i=1}^N \mathcal{L}(y_i, F_m(x_i) + \gamma f_{m+1}(x_i))$$

5. Update the model

$$F_{m+1}(x) = F_m(x) + \gamma_{m+1} f_{m+1}(x)$$

LightGBM uses leaf-wise tree growth, leaf with the highest loss reduction is split [3]

References



T. Hastie, R. Tibshirani, and J. Friedman.

The Elements of Statistical Learning.

Springer New York Inc., New York, NY, USA, 2001.



S. Lundberg and S.-I. Lee.

A unified approach to interpreting model predictions, 2017.



Y. Shi, G. Ke, D. Soukhavong, J. Lamb, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, T.-Y. Liu, N. Titov, and D. Cortes.

lightgbm: Light Gradient Boosting Machine, 2025.